Supplementary Tables

One-Sided Data Augmentation Pits Sensitivity against Invariance in Aspect-Based Sentiment Analysis

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Contents. These tables give the detailed inferential, diagnostic, and complete results that support the main article.

- Tables S1–S3 and S5–S13 are based on Study I (four encoders, two domains).
- Table S4 reports Study III audit-based sensitivity contrasts, including comparisons with Study II fine-tuned references.
- Tables S14–S16 report Study IV (LoRA fine-tuning of Qwen3-8B).

Conventions. Accuracy and Macro-F1 are percentages, and differences are percentage points (pp). Effects are measured against the exposure-matched Clean-Matched control unless a table states otherwise. Bracketed values are 95% source-cluster bootstrap intervals with trained models held fixed, so they quantify evaluation-source uncertainty; between-seed standard deviations (SD) are reported separately. Seeds are 13, 37, and 73 throughout.

Reproduction. Numerical records, study scripts, and verification commands for Studies I·IV are provided at github.com/saed-alqaraleh/absa-robustness-interference. The evidence index maps Tables S1·S16 to their corresponding study records and reconstruction instructions.

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Table S1: Inferential support for reciprocal REVTGT–REVNON interference (Study I)

Encoder	Domain	Direction	Δ (pp)	95% CI (pp)	p_{Holm}
BERT	Restaurant	REVTGT→REVNON	−24.40	[−29.04, −19.75]	0.0019
		REVNON→REVTGT	−34.99	[−38.42, −31.62]	0.0019
	Laptop	REVTGT→REVNON	−36.54	[−45.80, −27.07]	0.0019
		REVNON→REVTGT	−26.82	[−31.16, −22.42]	0.0019
RoBERTa	Restaurant	REVTGT→REVNON	−18.92	[−23.05, −14.95]	0.0019
		REVNON→REVTGT	−27.38	[−30.56, −24.22]	0.0019
	Laptop	REVTGT→REVNON	−24.20	[−31.60, −16.79]	0.0019
		REVNON→REVTGT	−23.75	[−27.60, −19.99]	0.0019
DeBERTa-v3	Restaurant	REVTGT→REVNON	−24.02	[−28.58, −19.69]	0.0019
		REVNON→REVTGT	−35.26	[−38.96, −31.71]	0.0019
	Laptop	REVTGT→REVNON	−47.41	[−56.04, −38.54]	0.0019
		REVNON→REVTGT	−24.46	[−28.10, −21.06]	0.0019
ModernBERT	Restaurant	REVTGT→REVNON	−16.74	[−20.44, −13.21]	< 0.001
		REVNON→REVTGT	−24.55	[−27.39, −21.82]	< 0.001
	Laptop	REVTGT→REVNON	−35.06	[−42.56, −27.60]	< 0.001
		REVNON→REVTGT	−20.53	[−24.17, −16.93]	< 0.001

Δ is the accuracy change relative to Clean-Matched; intervals are 5,000-replicate source-cluster bootstrap CIs. p_{Holm} is the paired sign-flip p-value after Holm adjustment: across 96 comparisons for BERT, RoBERTa, and DeBERTa-v3, and across the four prespecified comparisons for ModernBERT. Identical values reflect the Monte Carlo resolution floor of 50,000 permutations. Supports Section 5.1 and Table 6 of the main article.

Table S2: Edit-induced error relative to Clean-Matched (Study I, pp)

		REVTGT training → REVNON pairs			REVNON training → REVTGT pairs		
Encoder	Domain	Prior	ΔE	Favoured ΔE	Prior	ΔE	Favoured ΔE
BERT	Restaurant	+2.14	+25.48 [+20.7, +30.2]	+8.46 [−0.3, +16.7]	+0.21	+39.26 [+35.9, +42.7]	+7.69 [+3.1, +12.5]
	Laptop	−1.31	+40.52 [+31.0, +50.0]	+8.38 [+0.1, +18.9]	+0.37	+33.98 [+29.1, +39.0]	+7.45 [+1.2, +13.8]
RoBERTa	Restaurant	+3.04	+20.22 [+16.3, +24.1]	+3.04 [−2.5, +8.5]	−1.01	+30.65 [+27.4, +33.8]	+16.24 [+10.3, +22.4]
	Laptop	−0.63	+23.99 [+16.0, +32.0]	+8.27 [+0.0, +16.8]	+0.37	+28.62 [+24.4, +32.8]	+7.26 [+2.5, +12.5]
DeBERTa-v3	Restaurant	+0.80	+25.39 [+20.9, +29.9]	+8.18 [+3.7, +13.2]	+0.65	+40.09 [+36.6, +43.7]	+17.25 [+9.9, +24.9]
	Laptop	+6.32	+49.09 [+39.2, +58.2]	+5.59 [+1.8, +9.5]	−3.29	+28.68 [+24.4, +32.9]	+25.61 [+19.1, +32.6]
ModernBERT	Restaurant	+2.23	+16.52 [+12.8, +20.3]	+2.77 [−3.5, +8.6]	−0.15	+27.38 [+24.4, +30.3]	+13.07 [+7.8, +18.5]
	Laptop	+3.61	+37.76 [+30.0, +45.5]	+4.84 [−4.8, +13.8]	+3.08	+26.34 [+22.2, +30.4]	+16.23 [+9.5, +23.6]

ΔE is the change in edit-induced error (Eq. (7) of the main article): the rate at which a derivative is misclassified when its source is classified correctly. Prior is the change in the predicted share of the class favoured by the policy's training labels (negative for REVTGT training, positive for REVNON training) on unedited Clean reviews. Favoured ΔE restricts ΔE to pairs whose gold label is the favoured class: 66 and 26 REVNON pairs, and 142 and 100 REVTGT pairs, for Restaurant and Laptop. REVTGT pairs are label-changing. Brackets give 95% paired source-cluster bootstrap intervals with fixed seeds. Supports Section 5.2.

Table S3: Reversal co-exposure in RoBERTa: accuracy change relative to Clean-Matched (Study I, pp)

Domain	Evaluation	REVTGT-only	REVNON-only	REVTGT+REVNON (LOPO)	Mixed
Restaurant	REVTGT	+19.27 [+15.4, +23.3]	−27.38 [−30.5, −24.3]	+11.51 [+8.9, +14.2]	+9.50 [+6.6, +12.4]
	REVNON	−18.92 [−22.9, −14.9]	+5.56 [+3.8, +7.3]	+4.80 [+3.0, +6.6]	+4.05 [+2.3, +5.8]
	ADDDIFF	−5.86 [−7.5, −4.2]	+0.95 [−0.2, +2.1]	+1.16 [−0.03, +2.3]	+1.99 [+0.9, +3.1]
	Clean	−0.57 [−1.8, +0.7]	+0.86 [−0.1, +1.9]	+1.82 [+0.7, +2.9]	+1.07 [+0.1, +2.1]
Laptop	REVTGT	+13.30 [+9.3, +17.4]	−23.75 [−27.7, −20.0]	+2.43 [−1.0, +5.7]	+7.87 [+4.6, +11.2]
	REVNON	−24.20 [−31.6, −16.7]	+4.69 [−0.2, +10.1]	−9.88 [−15.0, −4.2]	+2.96 [−1.6, +7.2]
	ADDDIFF	+3.24 [+1.4, +5.2]	−0.94 [−3.0, +1.0]	−9.25 [−11.4, −7.0]	+6.17 [+4.3, +8.1]
	Clean	−0.99 [−2.4, +0.4]	−1.83 [−3.2, −0.5]	−4.55 [−6.4, −2.7]	−1.52 [−2.8, −0.2]

Results cover the three seeds shared with the transfer matrix. REVTGT+REVNON is the LOPO condition that divides the same budget B between the two reversal families and adds no insertion data. Brackets give 95% paired source-cluster bootstrap intervals with fixed seeds (5,000 replicates); they are resampled separately from those in Table S1, so shared contrasts can differ in the last digit. Supports Section 5.3.

Table S4: Audit-based exclusion sensitivity of the Study II and III contrasts (pp)

Contrast	Hyp.	Full cohorts	Invalid removed	Invalid and ambiguous removed
LLM zero-shot: W − Clean	H1	−11.1 [−14.1, −8.2]	−7.4 [−10.2, −4.8]	−3.8 [−6.3, −1.9]
LLM zero-shot: REVTGT − REVNON	H1	−11.5 [−15.3, −7.5]	−9.5 [−13.2, −5.8]	−6.8 [−10.4, −3.0]
LLM REVNON accuracy: REVTGT − Clean demonstrations	H2	+0.8 [0.0, +1.6]	+0.8 [0.0, +1.6]	+0.8 [0.0, +1.6]
LLM REVTGT accuracy: REVNON − Clean demonstrations	H2	+1.4 [+0.6, +2.1]	+1.4 [+0.6, +2.2]	+1.4 [+0.5, +2.2]
LLM J_{SI}: Mixed − Clean demonstrations	H3	+2.8 [+1.7, +3.9]	+3.1 [+1.8, +4.2]	+3.1 [+1.8, +4.4]
RoBERTa Paired − Mixed: W / J_{SI}		+1.7 / +4.0	+1.7 / +4.1	+1.8 / +4.3
RoBERTa Paired − LLM zero-shot: J_{SI}		+8.2 [+4.9, +11.5]	+8.6 [+5.1, +12.1]	+8.7 [+4.8, +12.5]

Removed items are those of the universally failed stratum (the census of items misclassified by all seven reference systems), using the author-verified verdicts: 48 Restaurant and 27 Laptop invalid items, and 83 and 59 invalid or ambiguous items. Brackets give 95% cluster-bootstrap intervals (10,000 replicates). Paired − Mixed values are point estimates on the reduced cohorts. The reduced cohorts are sensitivity analyses, not corrected benchmarks. Supports Section 5.7.

Table S5: Complete BERT results on Restaurant (Study I, mean ± SD, %)

	Accuracy				Macro-F1			
Training	Clean	RevTgt	RevNon	AddDiff	Clean	RevTgt	RevNon	AddDiff
Clean-Natural	$83.13_{\pm 0.27}$	$68.36_{\pm 1.57}$	$80.86_{\pm 2.88}$	$77.62_{\pm 0.72}$	$73.47_{\pm 1.10}$	$51.89_{\pm 1.72}$	$56.49_{\pm 4.36}$	$67.93_{\pm 1.82}$
Clean-Matched	$83.39_{\pm 0.32}$	$67.18_{\pm 0.25}$	$83.56_{\pm 1.79}$	$79.70_{\pm 2.27}$	$73.74_{\pm 1.20}$	$51.25_{\pm 1.78}$	$60.71_{\pm 1.53}$	$69.37_{\pm 3.07}$
Clean+RevTgt	$82.08_{\pm 0.42}$	$83.69_{\pm 0.71}$	$59.16_{\pm 5.48}$	$75.54_{\pm 6.64}$	$71.53_{\pm 0.93}$	$60.07_{\pm 4.04}$	$42.39_{\pm 2.76}$	$63.47_{\pm 8.87}$
Clean+RevNon	$83.33_{\pm 0.29}$	$32.19_{\pm 1.59}$	$89.86_{\pm 0.39}$	$79.08_{\pm 2.05}$	$73.93_{\pm 1.16}$	$32.79_{\pm 1.70}$	$67.45_{\pm 2.29}$	$68.65_{\pm 3.00}$
Clean+AddDiff	$83.78_{\pm 0.61}$	$66.27_{\pm 4.49}$	$82.28_{\pm 3.78}$	$83.78_{\pm 1.14}$	$74.99_{\pm 0.73}$	$51.71_{\pm 2.77}$	$59.65_{\pm 4.00}$	$68.25_{\pm 4.31}$
Clean+Mixed	$82.89_{\pm 0.73}$	$74.67_{\pm 1.10}$	$85.21_{\pm 2.34}$	$82.71_{\pm 1.36}$	$72.95_{\pm 1.67}$	$53.98_{\pm 4.25}$	$61.90_{\pm 1.00}$	$70.11_{\pm 3.06}$

Each cell gives the mean with its SD as a subscript, over seeds 13, 37, and 73. Clean-Natural adds no records; every other condition adds the same budget B to the shared clean base (Clean-Matched adds clean records).

Table S6: Complete BERT results on Laptop (Study I, mean ± SD, %)

	Accuracy				Macro-F1			
Training	Clean	RevTgt	RevNon	AddDiff	Clean	RevTgt	RevNon	AddDiff
Clean-Natural	$76.59_{\pm 0.65}$	$57.30_{\pm 2.95}$	$69.38_{\pm 8.68}$	$55.59_{\pm 8.70}$	$71.62_{\pm 1.75}$	$47.22_{\pm 2.13}$	$51.26_{\pm 4.79}$	$49.34_{\pm 5.85}$
Clean-Matched	$77.17_{\pm 0.48}$	$60.16_{\pm 1.08}$	$68.40_{\pm 10.64}$	$56.22_{\pm 0.74}$	$72.19_{\pm 1.18}$	$46.44_{\pm 1.38}$	$50.82_{\pm 4.95}$	$49.84_{\pm 1.81}$
Clean+RevTgt	$75.97_{\pm 1.57}$	$72.89_{\pm 2.83}$	$31.85_{\pm 5.59}$	$55.75_{\pm 16.50}$	$71.04_{\pm 1.60}$	$57.73_{\pm 3.90}$	$22.01_{\pm 3.67}$	$52.65_{\pm 13.05}$
Clean+RevNon	$73.77_{\pm 0.72}$	$33.33_{\pm 4.85}$	$77.78_{\pm 5.88}$	$46.66_{\pm 3.02}$	$67.71_{\pm 2.67}$	$29.99_{\pm 3.55}$	$56.77_{\pm 4.08}$	$38.11_{\pm 7.90}$
Clean+AddDiff	$74.56_{\pm 1.48}$	$59.01_{\pm 4.01}$	$54.07_{\pm 2.67}$	$70.90_{\pm 1.02}$	$68.29_{\pm 1.66}$	$44.15_{\pm 3.76}$	$41.00_{\pm 0.26}$	$59.00_{\pm 1.40}$
Clean+Mixed	$77.85_{\pm 0.36}$	$64.95_{\pm 0.66}$	$63.46_{\pm 3.73}$	$76.33_{\pm 0.00}$	$73.48_{\pm 0.25}$	$50.99_{\pm 0.76}$	$48.79_{\pm 5.21}$	$70.67_{\pm 0.93}$

Each cell gives the mean with its SD as a subscript, over seeds 13, 37, and 73. Clean-Natural adds no records; every other condition adds the same budget B to the shared clean base (Clean-Matched adds clean records).

Table S7: Complete RoBERTa results on Restaurant (Study I, mean ± SD, %)

	Accuracy				Macro-F1			
Training	Clean	RevTgt	RevNon	AddDiff	Clean	RevTgt	RevNon	AddDiff
Clean-Natural	$85.39_{\pm 0.95}$	$64.22_{\pm 6.94}$	$83.41_{\pm 2.75}$	$79.82_{\pm 1.44}$	$76.76_{\pm 1.82}$	$50.77_{\pm 6.71}$	$58.63_{\pm 1.72}$	$69.92_{\pm 1.38}$
Clean-Matched	$84.76_{\pm 1.42}$	$65.25_{\pm 8.13}$	$85.06_{\pm 0.47}$	$82.71_{\pm 2.69}$	$75.36_{\pm 2.44}$	$51.81_{\pm 7.25}$	$60.22_{\pm 3.99}$	$72.72_{\pm 4.56}$
Clean+RevTgt	$84.20_{\pm 3.65}$	$84.52_{\pm 1.86}$	$66.14_{\pm 2.39}$	$76.85_{\pm 7.99}$	$73.58_{\pm 9.08}$	$60.59_{\pm 8.56}$	$47.97_{\pm 7.45}$	$65.24_{\pm 13.18}$
Clean+RevNon	$85.62_{\pm 0.39}$	$37.86_{\pm 3.79}$	$90.62_{\pm 0.47}$	$83.66_{\pm 1.14}$	$77.00_{\pm 1.11}$	$37.41_{\pm 3.41}$	$68.92_{\pm 3.34}$	$74.74_{\pm 2.97}$
Clean+AddDiff	$86.10_{\pm 1.03}$	$67.93_{\pm 4.45}$	$82.96_{\pm 1.72}$	$83.54_{\pm 1.47}$	$78.07_{\pm 1.04}$	$52.31_{\pm 3.80}$	$60.28_{\pm 4.64}$	$67.75_{\pm 4.37}$
Clean+Mixed	$85.83_{\pm 1.30}$	$74.74_{\pm 2.15}$	$89.11_{\pm 0.34}$	$84.70_{\pm 1.00}$	$78.33_{\pm 1.98}$	$61.79_{\pm 1.87}$	$68.93_{\pm 2.46}$	$74.21_{\pm 2.13}$

Each cell gives the mean with its SD as a subscript, over seeds 13, 37, and 73. Clean-Natural adds no records; every other condition adds the same budget B to the shared clean base (Clean-Matched adds clean records).

Table S8: Complete RoBERTa results on Laptop (Study I, mean ± SD, %)

	Accuracy				Macro-F1			
Training	Clean	RevTgt	RevNon	AddDiff	Clean	RevTgt	RevNon	AddDiff
Clean-Natural	$77.64_{\pm 4.41}$	$56.72_{\pm 5.78}$	$68.15_{\pm 14.59}$	$53.66_{\pm 18.04}$	$71.31_{\pm 8.70}$	$46.07_{\pm 7.58}$	$51.72_{\pm 16.51}$	$47.84_{\pm 20.01}$
Clean-Matched	$81.87_{\pm 1.42}$	$61.30_{\pm 1.08}$	$82.96_{\pm 3.39}$	$72.78_{\pm 2.81}$	$78.77_{\pm 1.85}$	$50.25_{\pm 1.90}$	$71.50_{\pm 4.28}$	$68.30_{\pm 2.95}$
Clean+RevTgt	$80.88_{\pm 0.57}$	$74.61_{\pm 0.75}$	$58.77_{\pm 3.34}$	$76.02_{\pm 5.10}$	$77.49_{\pm 0.98}$	$60.90_{\pm 4.34}$	$49.16_{\pm 5.45}$	$71.92_{\pm 5.93}$
Clean+RevNon	$80.04_{\pm 0.86}$	$37.55_{\pm 2.89}$	$87.65_{\pm 2.60}$	$71.84_{\pm 1.48}$	$76.67_{\pm 1.78}$	$34.78_{\pm 1.37}$	$69.78_{\pm 1.89}$	$67.51_{\pm 1.94}$
Clean+AddDiff	$80.83_{\pm 1.07}$	$63.45_{\pm 3.68}$	$85.68_{\pm 3.34}$	$79.21_{\pm 0.96}$	$77.24_{\pm 1.49}$	$51.78_{\pm 2.29}$	$72.35_{\pm 3.34}$	$74.30_{\pm 0.64}$
Clean+Mixed	$80.36_{\pm 1.63}$	$69.17_{\pm 0.81}$	$85.93_{\pm 2.67}$	$78.94_{\pm 2.14}$	$76.59_{\pm 2.29}$	$53.03_{\pm 1.70}$	$68.24_{\pm 6.47}$	$74.57_{\pm 3.26}$

Each cell gives the mean with its SD as a subscript, over seeds 13, 37, and 73. Clean-Natural adds no records; every other condition adds the same budget B to the shared clean base (Clean-Matched adds clean records).

Table S9: Complete DeBERTa-v3 results on Restaurant (Study I, mean ± SD, %)

	Accuracy				Macro-F1			
Training	Clean	RevTgt	RevNon	AddDiff	Clean	RevTgt	RevNon	AddDiff
Clean-Natural	$86.19_{\pm 0.74}$	$70.72_{\pm 2.54}$	$84.53_{\pm 1.32}$	$80.92_{\pm 0.27}$	$77.33_{\pm 1.64}$	$53.89_{\pm 1.65}$	$59.19_{\pm 3.64}$	$69.96_{\pm 1.54}$
Clean-Matched	$86.10_{\pm 0.51}$	$67.85_{\pm 2.21}$	$86.64_{\pm 0.34}$	$82.20_{\pm 1.19}$	$77.45_{\pm 1.29}$	$51.36_{\pm 0.95}$	$62.06_{\pm 3.66}$	$71.16_{\pm 2.49}$
Clean+RevTgt	$84.17_{\pm 0.49}$	$85.11_{\pm 0.97}$	$62.61_{\pm 2.74}$	$80.36_{\pm 0.76}$	$73.81_{\pm 2.11}$	$59.94_{\pm 2.19}$	$45.19_{\pm 1.70}$	$66.75_{\pm 3.82}$
Clean+RevNon	$85.62_{\pm 1.17}$	$32.58_{\pm 4.75}$	$90.92_{\pm 0.26}$	$81.70_{\pm 0.62}$	$77.78_{\pm 1.78}$	$35.69_{\pm 4.74}$	$68.43_{\pm 1.72}$	$71.32_{\pm 1.68}$
Clean+AddDiff	$85.48_{\pm 1.38}$	$68.20_{\pm 1.63}$	$87.09_{\pm 0.34}$	$83.99_{\pm 0.99}$	$76.84_{\pm 3.17}$	$50.99_{\pm 2.27}$	$62.94_{\pm 2.65}$	$68.95_{\pm 3.88}$
Clean+Mixed	$84.94_{\pm 0.96}$	$79.55_{\pm 1.79}$	$89.11_{\pm 0.79}$	$83.27_{\pm 0.94}$	$75.02_{\pm 1.68}$	$60.27_{\pm 3.26}$	$63.19_{\pm 4.05}$	$68.99_{\pm 2.70}$

Each cell gives the mean with its SD as a subscript, over seeds 13, 37, and 73. Clean-Natural adds no records; every other condition adds the same budget B to the shared clean base (Clean-Matched adds clean records).

Table S10: Complete DeBERTa-v3 results on Laptop (Study I, mean ± SD, %)

	Accuracy				Macro-F1			
Training	Clean	RevTgt	RevNon	AddDiff	Clean	RevTgt	RevNon	AddDiff
Clean-Natural	$78.42_{\pm 0.80}$	$54.22_{\pm 9.23}$	$59.26_{\pm 10.37}$	$46.08_{\pm 9.40}$	$73.74_{\pm 1.44}$	$44.79_{\pm 5.93}$	$45.22_{\pm 11.47}$	$42.40_{\pm 10.20}$
Clean-Matched	$81.19_{\pm 2.00}$	$58.44_{\pm 6.49}$	$84.20_{\pm 0.86}$	$68.08_{\pm 5.52}$	$76.55_{\pm 2.63}$	$48.95_{\pm 4.72}$	$67.64_{\pm 6.64}$	$61.08_{\pm 7.05}$
Clean+RevTgt	$76.02_{\pm 6.46}$	$72.03_{\pm 5.84}$	$36.79_{\pm 9.67}$	$47.34_{\pm 19.96}$	$68.04_{\pm 13.09}$	$54.02_{\pm 8.22}$	$31.61_{\pm 10.47}$	$42.71_{\pm 20.72}$
Clean+RevNon	$77.38_{\pm 0.63}$	$33.98_{\pm 4.11}$	$78.77_{\pm 7.92}$	$45.25_{\pm 9.17}$	$72.33_{\pm 1.23}$	$28.06_{\pm 0.18}$	$55.30_{\pm 2.37}$	$40.60_{\pm 7.11}$
Clean+AddDiff	$79.94_{\pm 0.94}$	$59.66_{\pm 5.21}$	$78.77_{\pm 1.71}$	$76.23_{\pm 3.06}$	$75.60_{\pm 1.39}$	$47.98_{\pm 2.58}$	$58.14_{\pm 5.44}$	$68.48_{\pm 5.62}$
Clean+Mixed	$80.98_{\pm 1.16}$	$63.66_{\pm 3.69}$	$80.74_{\pm 9.01}$	$75.44_{\pm 1.81}$	$76.99_{\pm 1.61}$	$50.04_{\pm 1.61}$	$62.58_{\pm 6.53}$	$69.45_{\pm 1.79}$

Each cell gives the mean with its SD as a subscript, over seeds 13, 37, and 73. Clean-Natural adds no records; every other condition adds the same budget B to the shared clean base (Clean-Matched adds clean records).

Table S11: Complete ModernBERT results on Restaurant (Study I, mean ± SD, %)

	Accuracy				Macro-F1			
Training	Clean	RevTgt	RevNon	AddDiff	Clean	RevTgt	RevNon	AddDiff
Clean-Natural	$85.39_{\pm 0.21}$	$60.84_{\pm 2.13}$	$84.31_{\pm 1.30}$	$82.17_{\pm 2.67}$	$76.73_{\pm 0.58}$	$46.06_{\pm 2.32}$	$62.81_{\pm 1.76}$	$72.91_{\pm 4.27}$
Clean-Matched	$86.37_{\pm 0.22}$	$61.62_{\pm 3.24}$	$84.98_{\pm 2.37}$	$80.54_{\pm 2.36}$	$78.35_{\pm 0.59}$	$47.81_{\pm 1.06}$	$61.43_{\pm 5.86}$	$70.37_{\pm 4.05}$
Clean+RevTgt	$84.08_{\pm 1.00}$	$84.28_{\pm 0.83}$	$68.24_{\pm 6.35}$	$77.98_{\pm 0.67}$	$73.73_{\pm 3.54}$	$61.02_{\pm 1.03}$	$47.24_{\pm 7.80}$	$63.48_{\pm 5.25}$
Clean+RevNon	$83.57_{\pm 1.52}$	$37.08_{\pm 5.43}$	$89.71_{\pm 1.69}$	$78.12_{\pm 3.48}$	$73.84_{\pm 2.11}$	$33.48_{\pm 4.38}$	$64.15_{\pm 2.99}$	$65.56_{\pm 4.83}$
Clean+AddDiff	$84.64_{\pm 0.54}$	$57.49_{\pm 1.18}$	$82.73_{\pm 2.85}$	$82.38_{\pm 0.21}$	$74.94_{\pm 1.43}$	$42.90_{\pm 0.78}$	$54.40_{\pm 2.76}$	$64.17_{\pm 1.27}$
Clean+Mixed	$84.32_{\pm 1.28}$	$72.14_{\pm 4.07}$	$88.66_{\pm 1.13}$	$82.62_{\pm 2.78}$	$74.74_{\pm 2.22}$	$53.19_{\pm 6.41}$	$62.85_{\pm 4.07}$	$68.63_{\pm 6.00}$

Each cell gives the mean with its SD as a subscript, over seeds 13, 37, and 73. Clean-Natural adds no records; every other condition adds the same budget B to the shared clean base (Clean-Matched adds clean records).

Table S12: Complete ModernBERT results on Laptop (Study I, mean ± SD, %)

	Accuracy				Macro-F1			
Training	Clean	RevTgt	RevNon	AddDiff	Clean	RevTgt	RevNon	AddDiff
Clean-Natural	$78.84_{\pm 1.81}$	$52.43_{\pm 7.42}$	$66.67_{\pm 12.64}$	$57.84_{\pm 13.57}$	$74.99_{\pm 2.38}$	$43.96_{\pm 5.20}$	$50.95_{\pm 8.91}$	$54.31_{\pm 12.79}$
Clean-Matched	$78.37_{\pm 0.98}$	$54.29_{\pm 4.99}$	$74.32_{\pm 3.73}$	$65.05_{\pm 2.76}$	$73.87_{\pm 1.18}$	$44.23_{\pm 3.03}$	$52.37_{\pm 5.65}$	$57.86_{\pm 3.63}$
Clean+RevTgt	$76.54_{\pm 2.67}$	$73.18_{\pm 1.97}$	$39.26_{\pm 11.64}$	$51.41_{\pm 15.47}$	$72.75_{\pm 3.16}$	$56.68_{\pm 3.16}$	$29.96_{\pm 8.75}$	$49.99_{\pm 13.61}$
Clean+RevNon	$73.25_{\pm 3.41}$	$33.76_{\pm 3.96}$	$80.74_{\pm 6.46}$	$51.99_{\pm 10.92}$	$66.15_{\pm 5.46}$	$28.99_{\pm 1.91}$	$57.53_{\pm 5.18}$	$43.31_{\pm 10.76}$
Clean+AddDiff	$79.36_{\pm 0.48}$	$58.01_{\pm 2.17}$	$71.60_{\pm 7.46}$	$76.59_{\pm 2.17}$	$75.30_{\pm 1.22}$	$46.68_{\pm 1.37}$	$50.72_{\pm 4.31}$	$68.70_{\pm 4.20}$
Clean+Mixed	$78.74_{\pm 0.09}$	$62.59_{\pm 3.44}$	$76.30_{\pm 4.86}$	$74.66_{\pm 2.40}$	$74.54_{\pm 0.53}$	$47.00_{\pm 1.88}$	$55.63_{\pm 1.76}$	$66.55_{\pm 4.27}$

Each cell gives the mean with its SD as a subscript, over seeds 13, 37, and 73. Clean-Natural adds no records; every other condition adds the same budget B to the shared clean base (Clean-Matched adds clean records).

Table S13: Reciprocal REVTGT–REVNON effects relative to Clean-Natural (Study I, pp)

		Relative to Clean-Natural		Clean-Matched − Clean-Natural	
Encoder	Domain	REVTGT→REVNON	REVNON→REVTGT	on REVNON	on REVTGT
BERT	Restaurant	−21.70	−36.17	+2.70	−1.18
	Laptop	−37.53	−23.97	−0.98	+2.86
RoBERTa	Restaurant	−17.27	−26.36	+1.65	+1.03
	Laptop	−9.38	−19.17	+14.81	+4.58
DeBERTa-v3	Restaurant	−21.92	−38.14	+2.11	−2.87
	Laptop	−22.47	−20.24	+24.94	+4.22
ModernBERT	Restaurant	−16.07	−23.76	+0.67	+0.78
	Laptop	−27.41	−18.67	+7.65	+1.86

Differences of seed-mean accuracies in Tables S5–S12. REVTGT→REVNON is REVNON accuracy after REVTGT training minus REVNON accuracy of Clean-Natural; REVNON→REVTGT is defined analogously. All 16 effects are negative. The last two columns give the shift in the reference produced by matched clean exposure; effects relative to Clean-Matched (Table S1) equal the first two columns minus these shifts, up to rounding. Supports Section 5.1.

Table S14: Study IV transfer matrix: LoRA fine-tuning of Qwen3-8B relative to Clean-Matched (pp)

Domain	Training	Evaluation family		
		RevTgt	RevNon	AddDiff
Restaurant	Clean+RevTgt	+9.50 [+6.62, +12.56] $p < 0.001$	−2.33 [−4.16, −0.53] $p = 0.0126$	−0.80 [−1.71, +0.06] $p = 0.0786$
	Clean+RevNon	−43.66 [−47.36, −39.94] $p < 0.001$	+4.88 [+2.64, +7.33] $p < 0.001$	−0.60 [−1.47, +0.24] $p = 0.1816$
	Clean+Mixed	+2.72 [+0.54, +4.86] $p = 0.0138$	+2.93 [+1.47, +4.48] $p < 0.001$	−0.65 [−1.61, +0.25] $p = 0.1700$
Laptop	Clean+RevTgt	+4.01 [+1.79, +6.20] $p < 0.001$	+1.73 [−0.55, +4.17] $p = 0.1660$	−1.04 [−2.20, +0.11] $p = 0.0804$
	Clean+RevNon	−33.33 [−37.96, −28.74] $p < 0.001$	+4.20 [+0.72, +7.94] $p = 0.0180$	−0.78 [−1.85, +0.25] $p = 0.1452$
	Clean+Mixed	−1.29 [−3.95, +1.03] $p = 0.3300$	+2.47 [0.00, +5.13] $p = 0.0518$	−0.16 [−1.06, +0.72] $p = 0.7678$

T_{ij} is the seed mean of seed-matched differences from Clean-Matched (seeds 13, 37, and 73). Brackets give unadjusted 95% percentile intervals from 10,000 resamples of ARTS sentence clusters (seed 20261007), with the same resample applied to every run, so all contrasts are paired. p is the two-sided bootstrap p-value, shown as $p < 0.001$ below 0.001 (the resolution floor is $1/10{,}000$). The four prespecified L1 contrasts, with Holm adjustment, appear in Table S15. Supports Section 5.6 and Table 11.

Table S15: Study IV prespecified hypotheses: seed-level L1 contrasts and the L2 co-exposure contrasts (pp)

(a) L1: reciprocal interference, relative to Clean-Matched

Domain	Direction	Seed 13	Seed 37	Seed 73	Mean	Holm-adjusted 95% CI	p_{Holm}
Restaurant	RevTgt→RevNon	−5.86	−0.45	−0.68	−2.33	[−4.46, −0.24] (97.50%)	0.0252
Restaurant	RevNon→RevTgt	−45.51	−42.20	−43.26	−43.66	[−48.30, −38.91] (98.75%)	< 0.001
Laptop	RevTgt→RevNon	0.00	+2.96	+2.22	+1.73	[−0.55, +4.17] (95.00%)	0.1660
Laptop	RevNon→RevTgt	−31.33	−30.47	−38.20	−33.33	[−39.18, −27.73] (98.33%)	< 0.001

(b) L2: Clean+Mixed − Clean-Matched

Endpoint	Equal-domain mean	p	Restaurant	Laptop
R_{mean}	+1.00 [+0.19, +1.83]	0.0138	+1.66 [+0.71, +2.64]	+0.34 [−0.93, +1.64]
W	+0.72 [−0.99, +2.33]	0.3832	+2.72 [+0.54, +4.86]	−1.29 [−3.95, +1.03]
J_{SI}	+6.35 [+4.46, +8.37]	< 0.001	+8.26 [+5.87, +10.89]	+4.44 [+1.57, +7.58]

(a) The interval for the contrast ranked r (0-based) by its unadjusted p-value is the percentile interval at level $1 - 0.05/(4 - r)$, shown in parentheses. The reading fixed in advance is "reproduced" if all four means are negative and all four adjusted intervals exclude zero, "partly reproduced" if all four means are negative, and "not reproduced" otherwise. The observed reading is *not reproduced*: 9 of the 12 seed-level contrasts are negative, 1 is zero, and 2 are positive. (b) Unadjusted 95% intervals; domains are resampled independently and averaged with equal weight. The L2 reading is *partly supported*. Both panels use the resampling scheme of Table S14.

Table S16: Complete Study IV profiles: Qwen3-8B zero-shot and LoRA fine-tuned (mean ± SD, %)

Domain	Training	Clean	REVTGT	REVNON	ADDDIFF	R_{mean}	W	J_{SI}
Restaurant	Zero-shot	86.43	76.00	80.63	86.16	80.93	76.00	66.44
	Clean-Matched	$90.74_{\pm 0.54}$	$76.87_{\pm 1.81}$	$87.76_{\pm 0.69}$	$88.72_{\pm 0.22}$	$84.45_{\pm 0.89}$	$76.87_{\pm 1.81}$	$71.85_{\pm 1.62}$
	Clean+REVTGT	$89.97_{\pm 0.52}$	$86.37_{\pm 0.89}$	$85.44_{\pm 3.62}$	$87.92_{\pm 0.26}$	$86.57_{\pm 1.15}$	$84.51_{\pm 2.88}$	$78.45_{\pm 2.47}$
	Clean+REVNON	$89.67_{\pm 0.22}$	$33.22_{\pm 3.29}$	$92.64_{\pm 1.28}$	$88.12_{\pm 0.39}$	$71.33_{\pm 0.91}$	$33.22_{\pm 3.29}$	$33.26_{\pm 4.03}$
	Clean+Mixed	$89.97_{\pm 0.95}$	$79.59_{\pm 1.25}$	$90.69_{\pm 0.47}$	$88.07_{\pm 0.10}$	$86.12_{\pm 0.27}$	$79.59_{\pm 1.25}$	$80.11_{\pm 0.13}$
Laptop	Zero-shot	80.25	71.89	84.44	79.78	78.70	71.89	63.70
	Clean-Matched	$83.28_{\pm 1.26}$	$73.03_{\pm 0.50}$	$84.94_{\pm 0.43}$	$82.97_{\pm 0.39}$	$80.31_{\pm 0.13}$	$73.03_{\pm 0.50}$	$69.63_{\pm 0.74}$
	Clean+REVTGT	$82.55_{\pm 1.01}$	$77.04_{\pm 0.64}$	$86.67_{\pm 1.48}$	$81.92_{\pm 0.86}$	$81.88_{\pm 0.38}$	$77.04_{\pm 0.64}$	$73.83_{\pm 2.60}$
	Clean+REVNON	$83.44_{\pm 1.18}$	$39.70_{\pm 3.74}$	$89.14_{\pm 0.43}$	$82.18_{\pm 1.01}$	$70.34_{\pm 0.89}$	$39.70_{\pm 3.74}$	$40.00_{\pm 4.44}$
	Clean+Mixed	$83.44_{\pm 0.50}$	$71.75_{\pm 1.61}$	$87.41_{\pm 0.74}$	$82.81_{\pm 0.45}$	$80.65_{\pm 0.76}$	$71.75_{\pm 1.61}$	$74.07_{\pm 1.96}$

Each cell gives the mean with its SD as a subscript, over seeds 13, 37, and 73; the zero-shot reference is one deterministic job. R_{mean} is the mean of the three ARTS family accuracies, and W is the worst-family accuracy computed per run before averaging. J_{SI} uses the 444 Restaurant and 135 Laptop sources with both REVTGT and REVNON items. Every condition adds the same Study I budget B to the shared clean base; the model, prompt, recipe, and pinned revision are described in Sections 3.7 and 4.7.